Statistical Po

Overview

1. **The Central Limit Theorem revisited**
2. **From Null worlds to True effect worlds**
3. **Statistical power**
4. **A power simulation**

The Central Limit Theorem revisited

Are action movies better than

In the previous session on hypothesis testing, we invented a

We simulated a population of 1’000’000 movies with no diff

```
set.seed(1234) # For reproducibility

imaginary_movies_null <- tibble(
  movie_id = 1:1000000,
  rating = sample(seq(1, 10, by = 0.1), size = 1000000, replace = TRUE),
  genre = sample(c("Comedy", "Action"), size = 1000000, replace = TRUE)
)
```

Sampling distribution

We randomly drew 1,000 samples from this population and
between action movies and comedies for each.

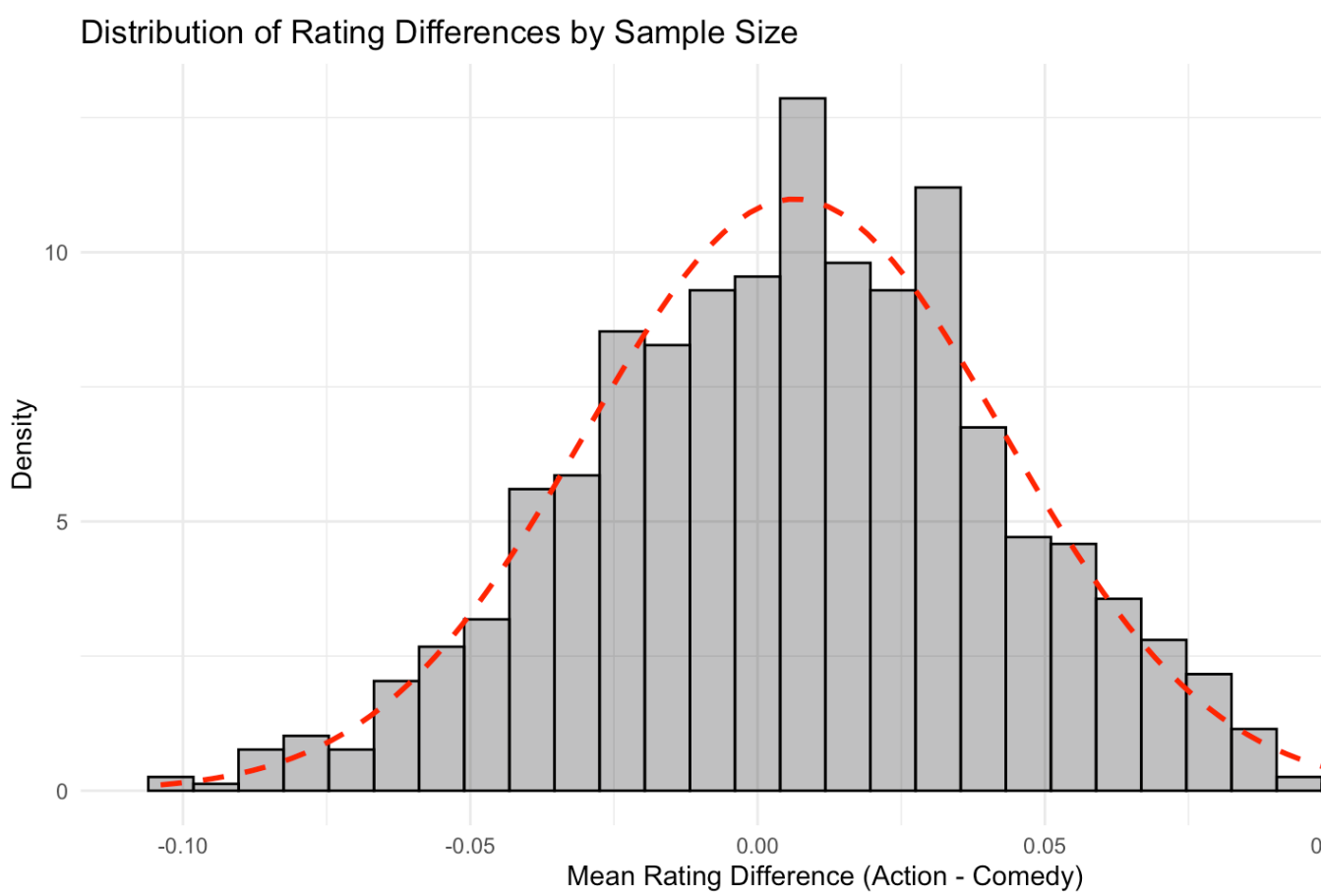
We called the distribution of the differences from the differ
distribution

Our sample size was always the same: 20,000.

```
n_simulations <- 1000
differences <- c() # make an empty vector
sample_size <- 20000

for (i in 1:n_simulations) {
  # draw a sample of 20'000 films
  imaginary_sample <- imaginary_movies |>
    sample_n(sample_size)
  # compute rating difference in the sample
  estimate <- imaginary_sample |>
    group_by(genre) |>
    summarize(avg_rating = mean(rating)) |>
    summarise(diff = avg_rating[genre == "Action"] - avg_rating[genre ==
    pull(diff)

  differences[i] <- estimate
}
```

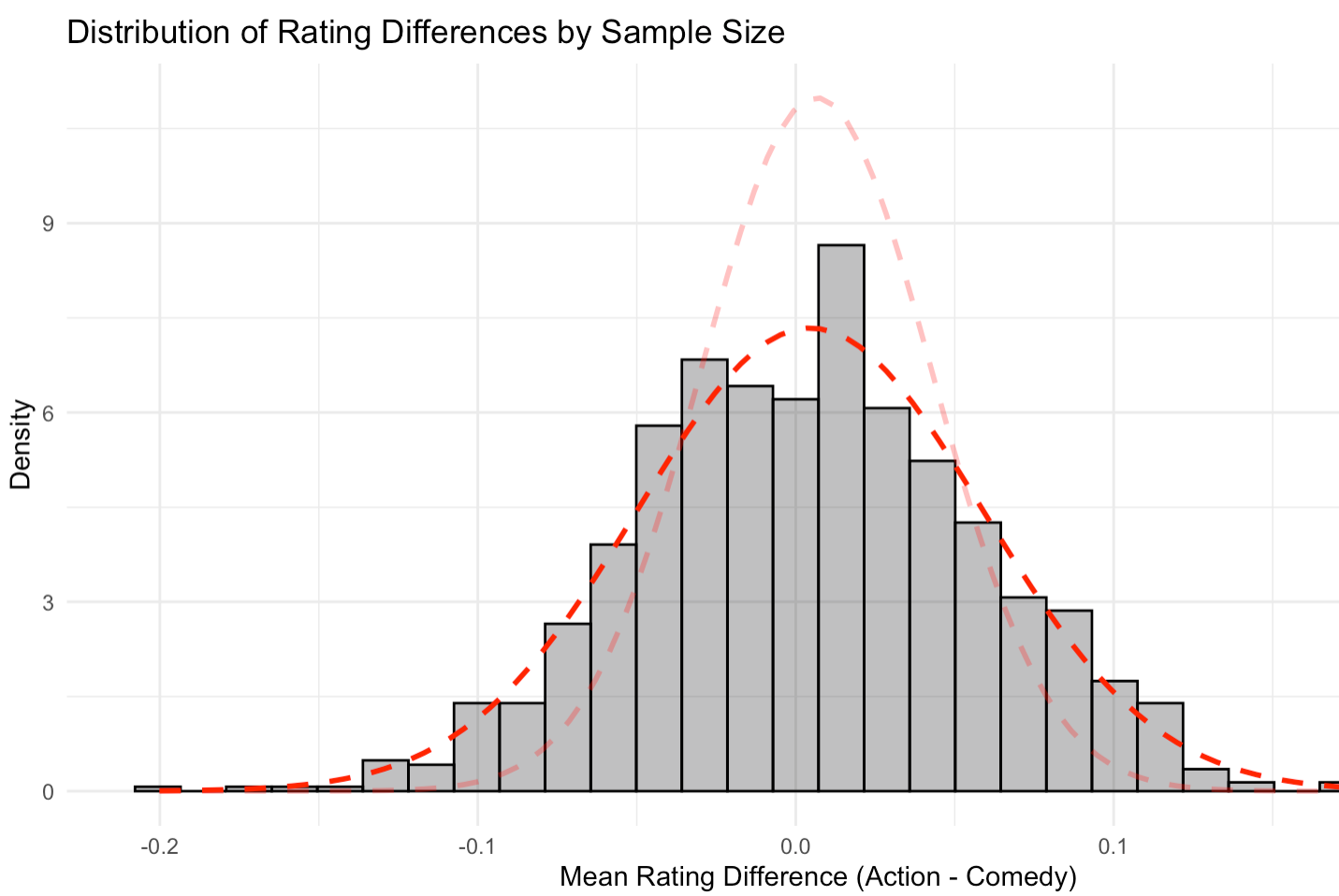
The Central Limit Theorem (

The sampling distribution approximates the shape of a norm



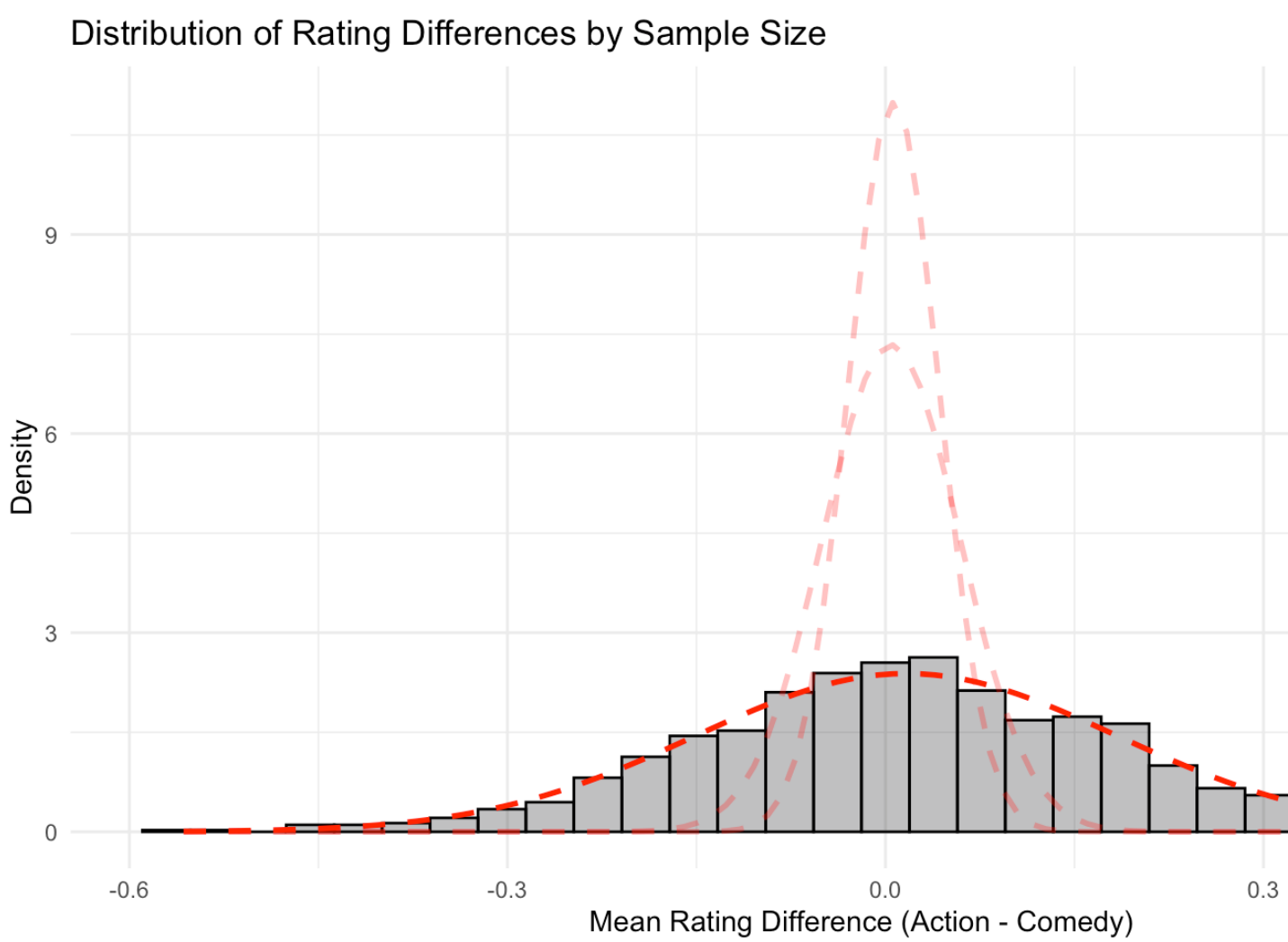
The Central Limit Theorem (

This is what the sampling distribution looks like with sample



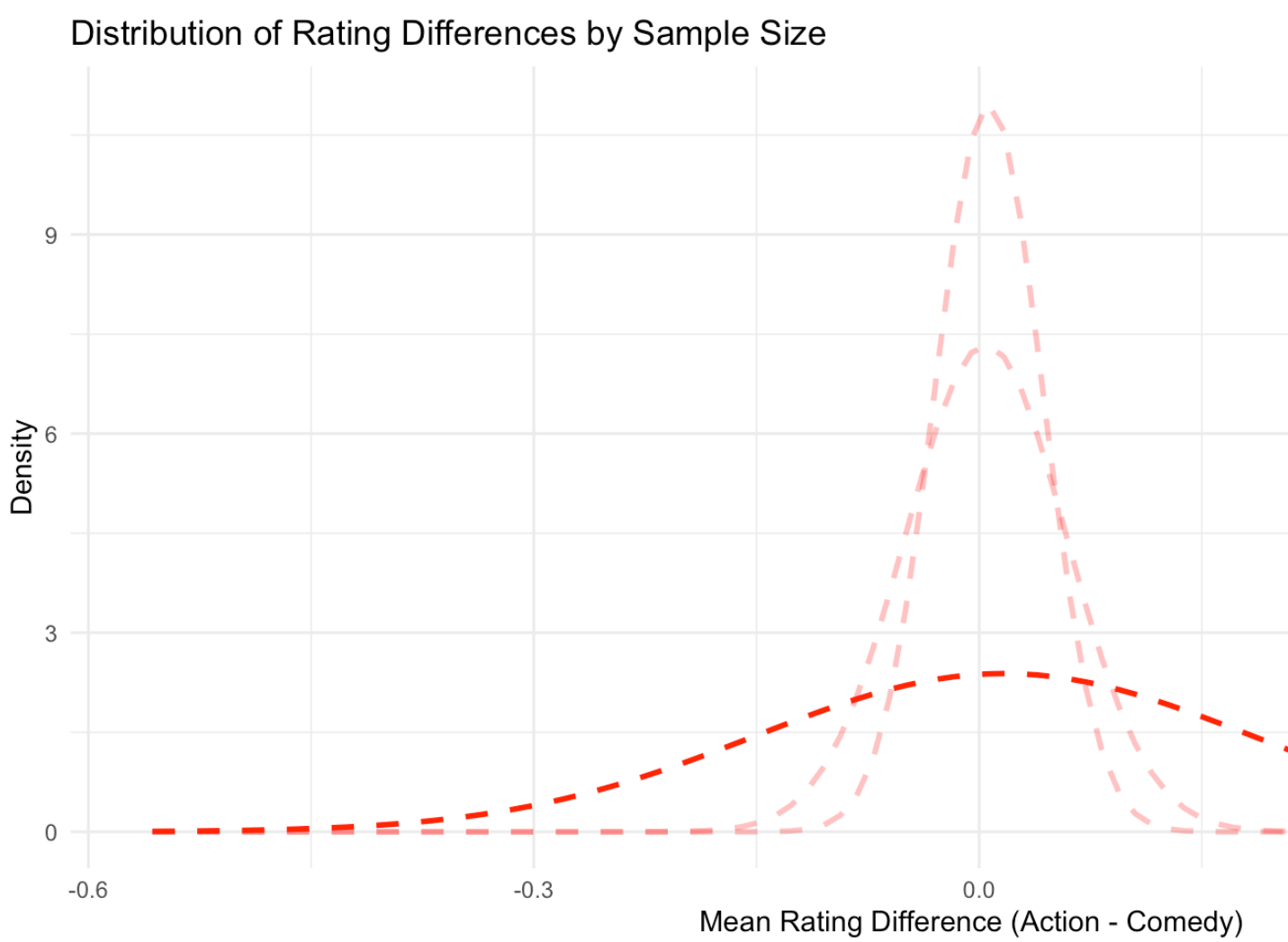
The Central Limit Theorem (

And with samples of size 1,000



The Central Limit Theorem (

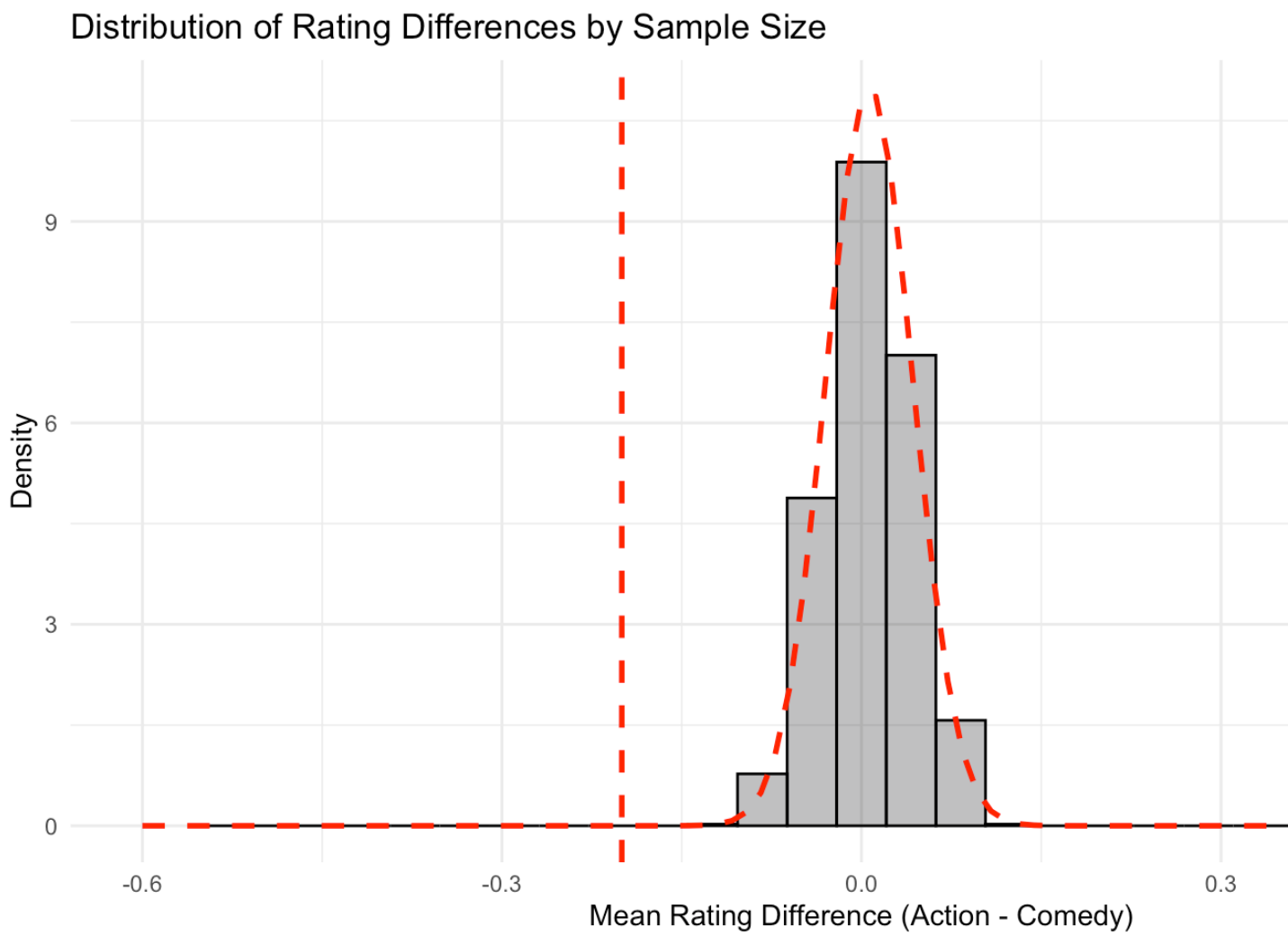
The smaller the sample, the larger the standard deviation of



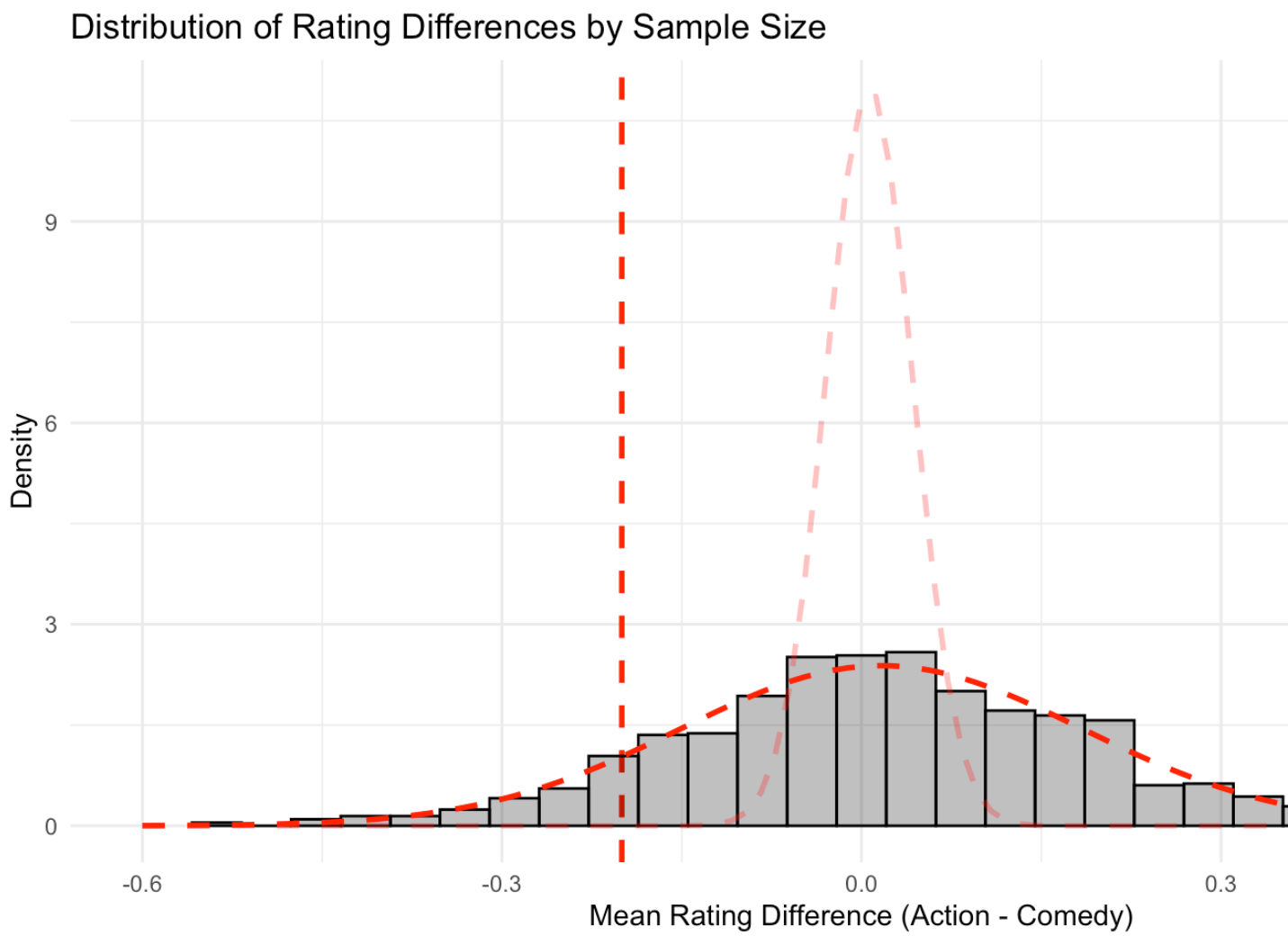
How is this relevant for hypothesis testing?

Imagine we find an effect of -0.2 in our sample

In a sample based on 20’000 movies, that seems reasonably



But the same effect in a sample of 1000 seems not so unlike



The Central Limit Theorem

(part I)

- The sampling distribution approximates the shape of a no

(part II)

- The smaller the sample, the larger the standard deviation
  distribution

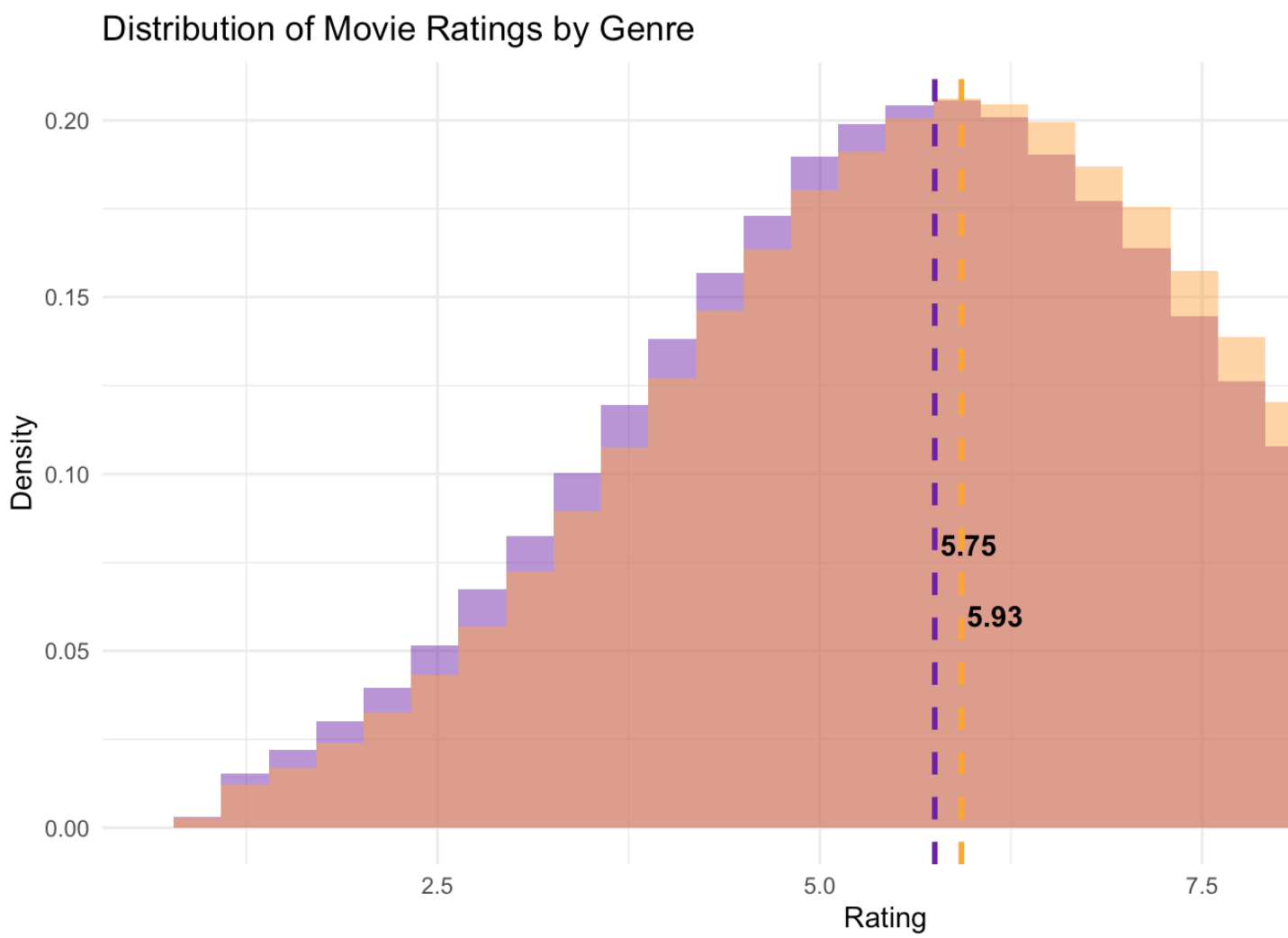
From Null worlds t
effect worlds

For example, let's imagine a world where the true difference
comedy movies is -0.2

```
# Generate Comedy and Action movie ratings using truncated normal distrib
imaginary_movies_true <- tibble(
  movie_id = 1:1000000,
  genre = sample(c("Comedy", "Action"), size = 1000000, replace = TRUE),
  rating = ifelse(
    genre == "Comedy",
    rtruncnorm(1000000, a = 1, b = 10, mean = 6.0, sd = 2),
    rtruncnorm(1000000, a = 1, b = 10, mean = 5.8, sd = 2)
  )
)
```

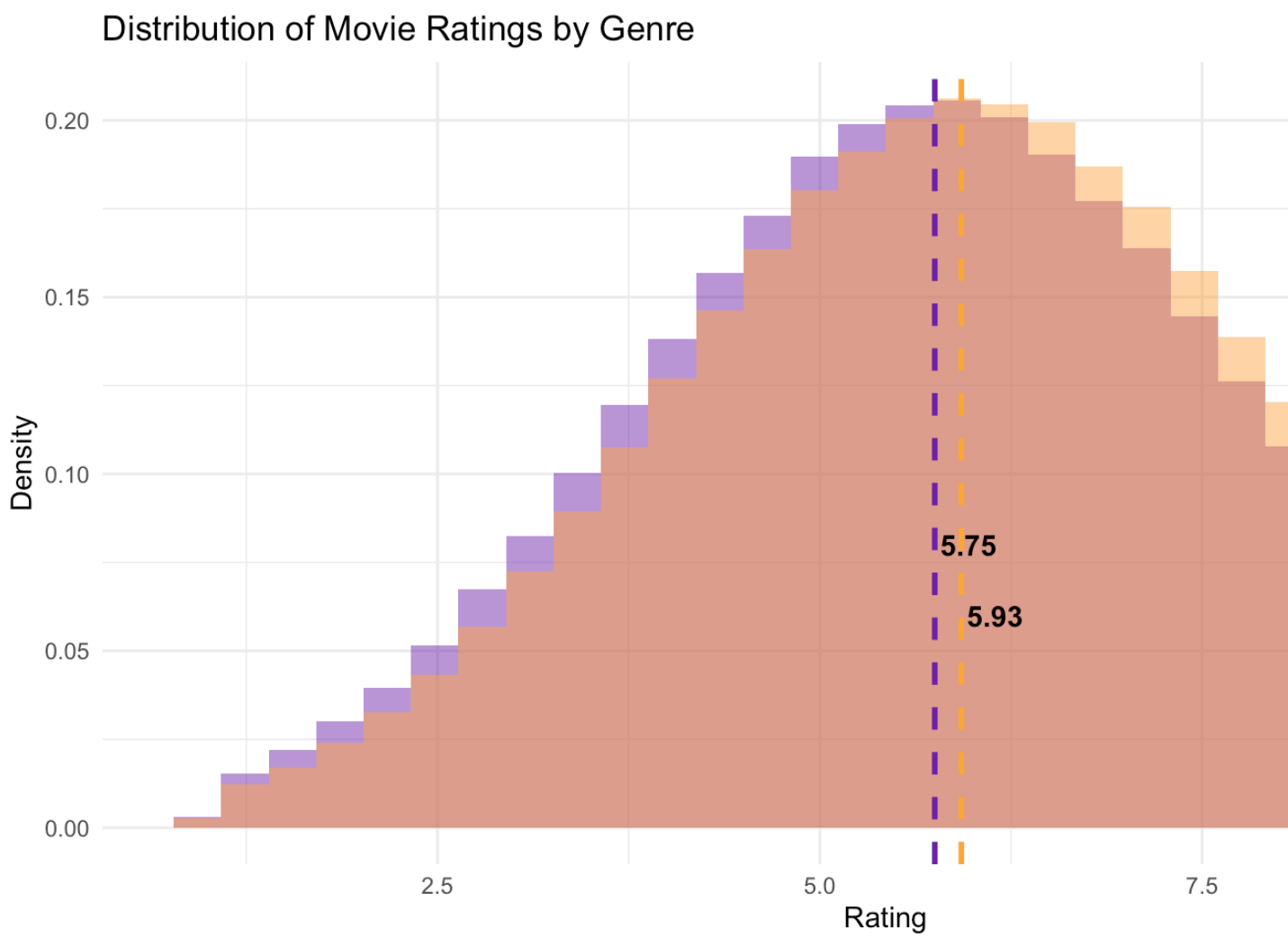
Let's plot the population data

- Code



🎉It worked, we see our imagined effect of ~ -0.2

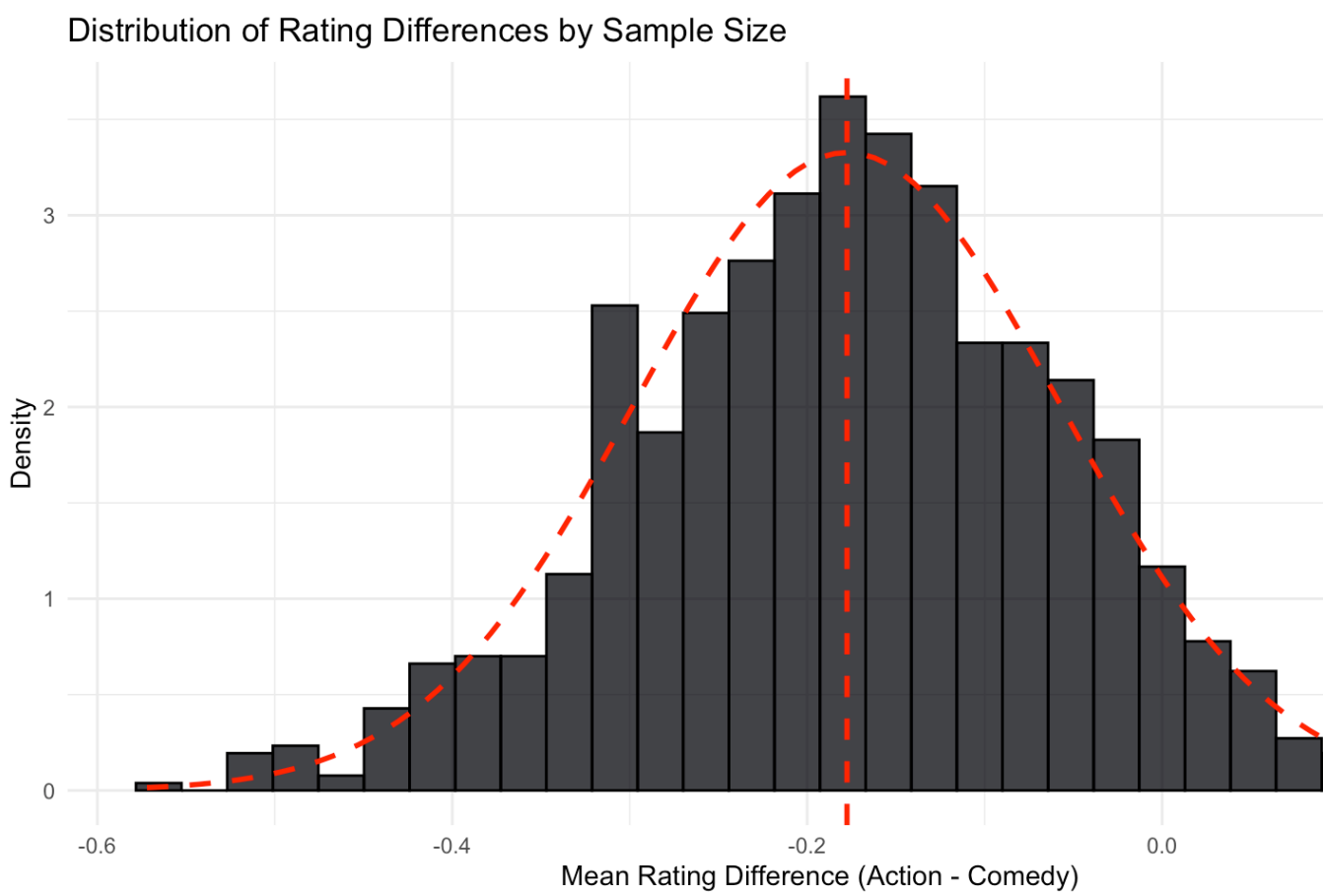
- Code



Imagine we look at a sample of 1000 movie ratings.

```
n_simulations <- 1000
differences <- c() # make an empty vector
sample_size <- 1000

for (i in 1:n_simulations) {
  # draw a sample of 20'000 films
  imaginary_sample <- imaginary_movies |>
    sample_n(sample_size)
  # compute rating difference in the sample
  estimate <- imaginary_sample |>
    group_by(genre) |>
    summarize(avg_rating = mean(rating)) |>
    summarise(diff = avg_rating[genre == "Action"] - avg_rating[genre ==
    pull(diff)

  differences[i] <- estimate
}
```

This is what our sampling distribution would look like



Now imagine we do a hypothesis test.

We can simulate both a null world and a true effect world (s

Null world

```
# Null world population
imaginary_movies_null <- tibble(
  movie_id = 1:1000000,
  rating = sample(seq(1, 10, by = 0.1), size = 100
  genre = sample(c("Comedy", "Action"), size = 100
)
```

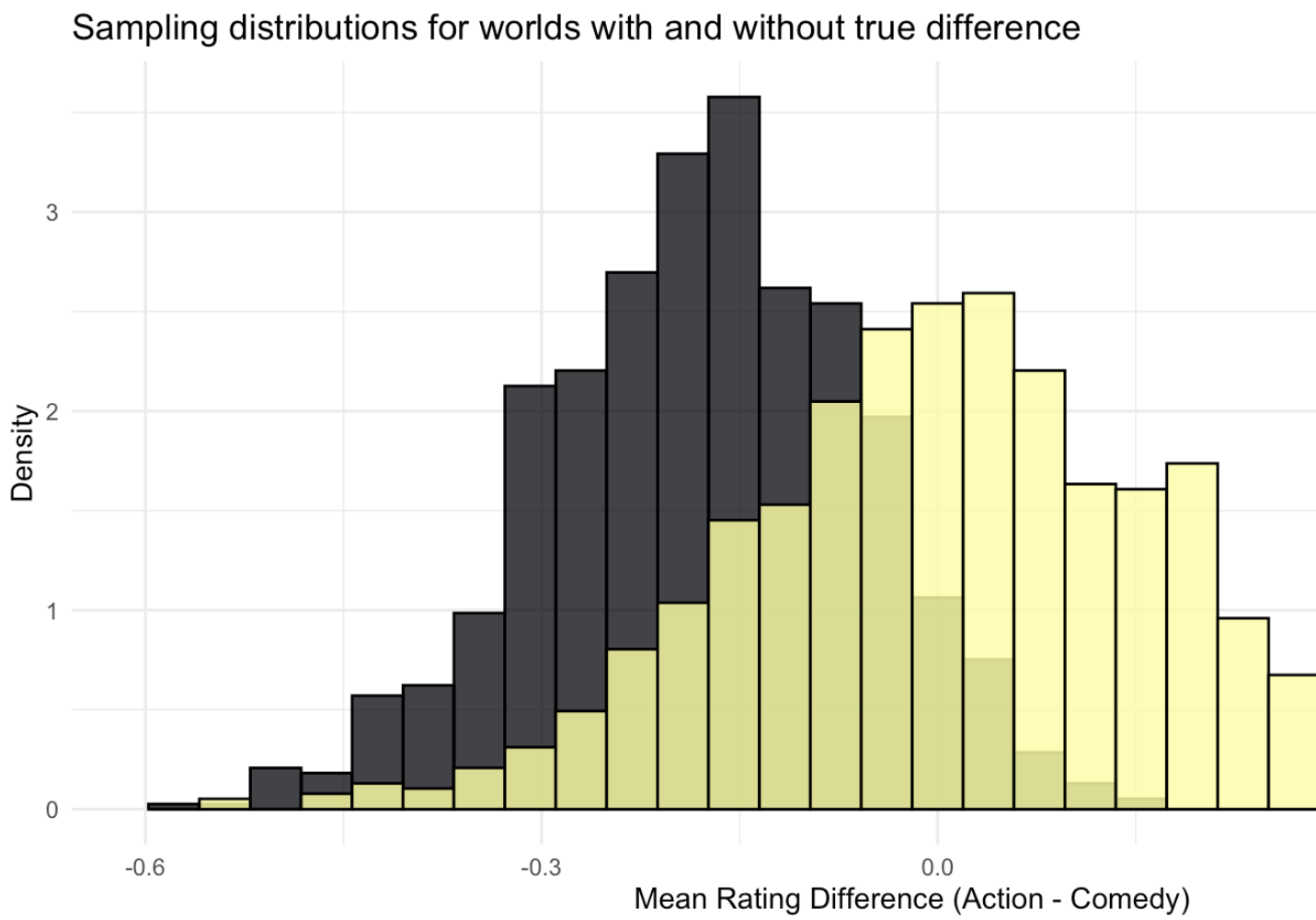
True effect world

```
# True effect worl
imaginary_movies_t
  movie_id = 1:100
  genre = sample(c
  rating = ifelse(
    genre == "Come
    rtruncnorm(100
    rtruncnorm(100
  )
)
```

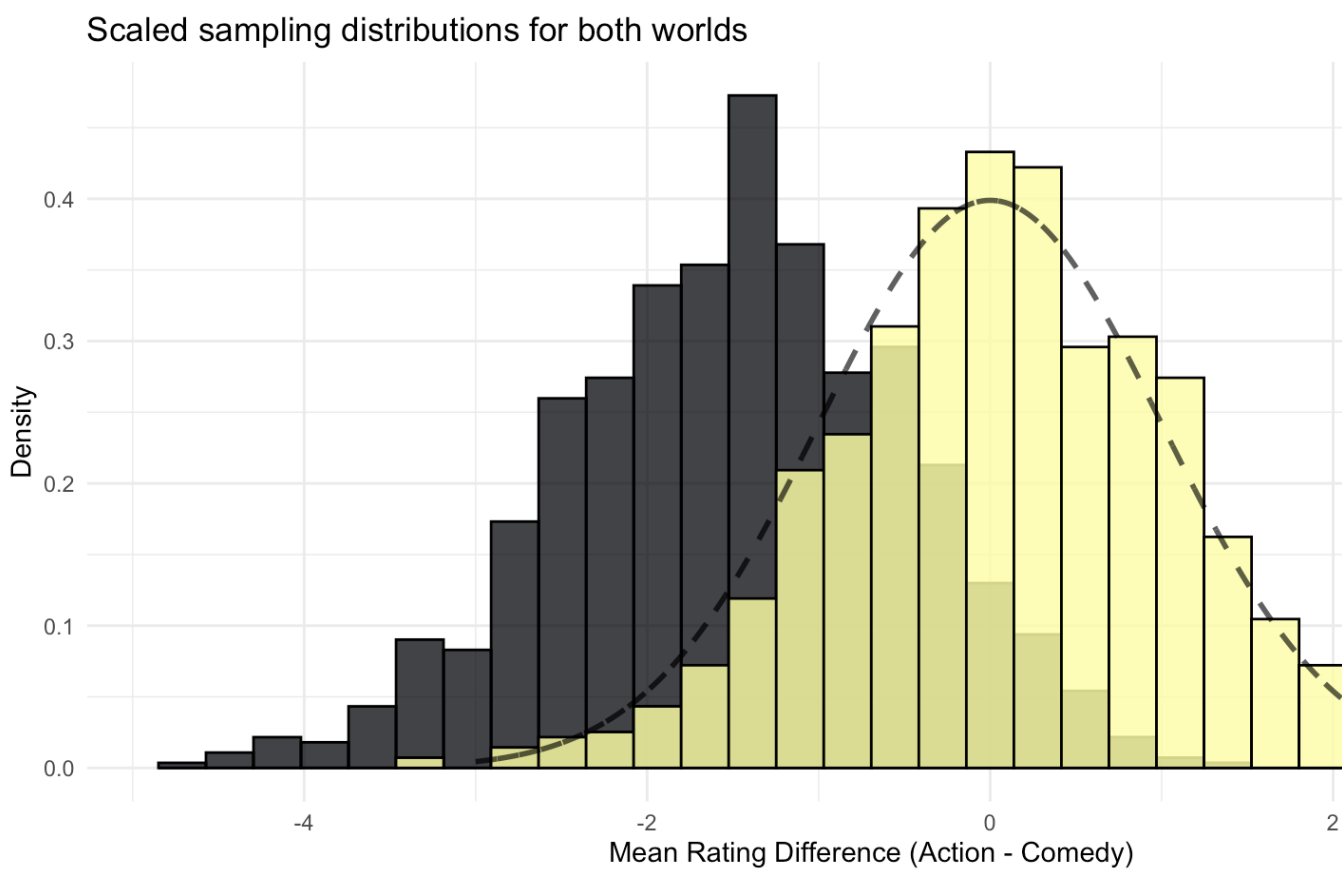
And make a sampling distribution with sample size 1000 for new)

```
n_simulations <- 1000
differences <- c() # make an empty vector
sample_size <- 1000

data <- imaginary_movies_true # replace with imaginary_movies_null

for (i in 1:n_simulations) {
  # draw a sample of films
  imaginary_sample <- data |>
    sample_n(sample_size)
  # compute rating difference in the sample
  estimate <- imaginary_sample |>
    group_by(genre) |>
    summarize(avg_rating = mean(rating)) |>
    summarise(diff = avg_rating[genre == "Action"] - avg_rating[genre ==
    pull(diff)

  differences[i] <- estimate
}
```

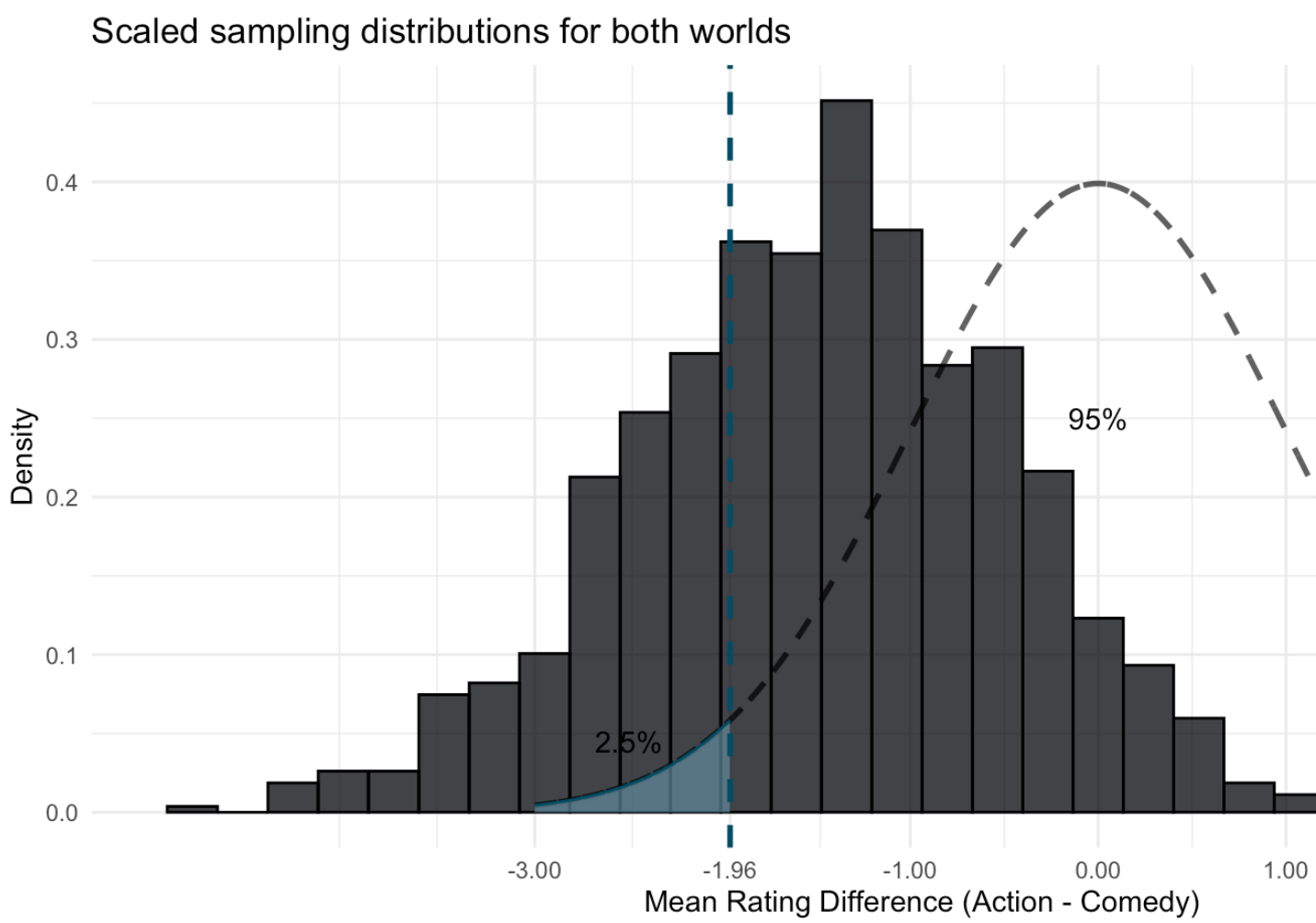
We can plot both simulated worlds together.



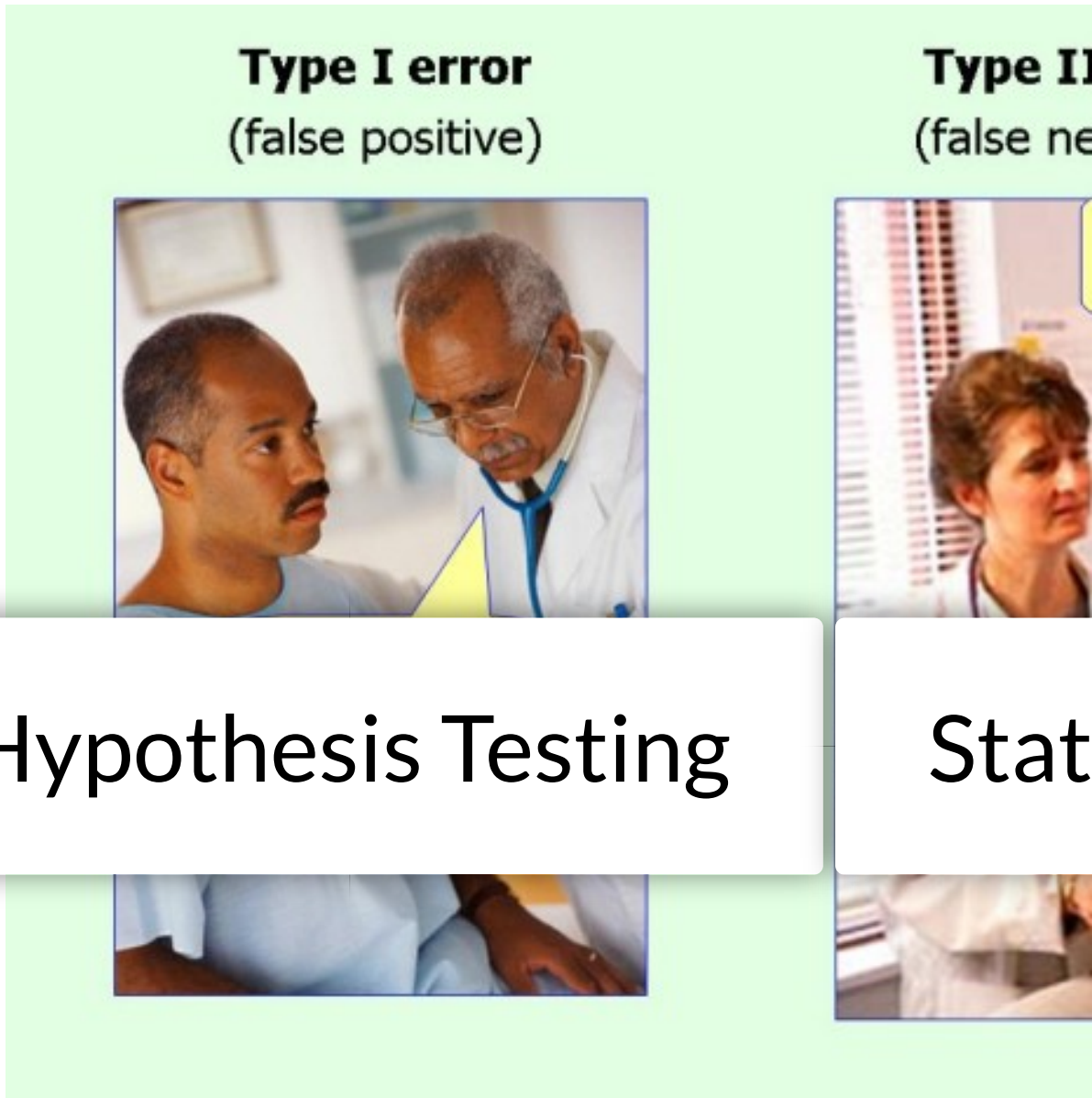
Let’s bring both worlds on the scale of a standard normal dis
respective standard deviation



With a sample size of 1000, in many cases, we would say: “T
a Null World”



Two errors



Hypothesis Testing

Stat

Two errors

	Hypothesis Testing	**Power Ana**
Aim	Rule out that we observe something just by chance.	Ensure tha
Error question	"What are the chances that we find an effect at least this large in our sample, given that there is **no effect** in the population?"	"What are **not find a s effect in ou is a certain population**
Typical threshold for acceptable error	α = 5 %	β = 20 %

Statistical Power

Statistical Power

Statistical power is the probability of detecting an effect wit
certain effect size

(Or $1 - \beta$)

Your turn #1: Calculating pow

1. Create your true effect world: Simulate a population with
   between action and comedy movies.
2. Get your sampling distribution: Simulate 1000 random sa
   store the results.
3. Plot your Sampling distribution (use `data.frame()` to t
   frame that can be read by `ggplot`)
4. Prepare for hypothesis testing: bring the results on scale
   distribution (divide by the standard deviation of your dist
5. Check the transformation: plot the new, standardized val
6. Calculate your power: Check how many (standardized) di
   threshold, i.e. smaller than or equal to -1.96 (hint: use `mut`
   with `ifelse` to create a new variable `significant` tha
   or `FALSE`. Then use `summarize()` and `sum()` to calcula
   above the 80% power threshold?

1. Create your true effect world: Simulate a population with
   between action and comedy movies.

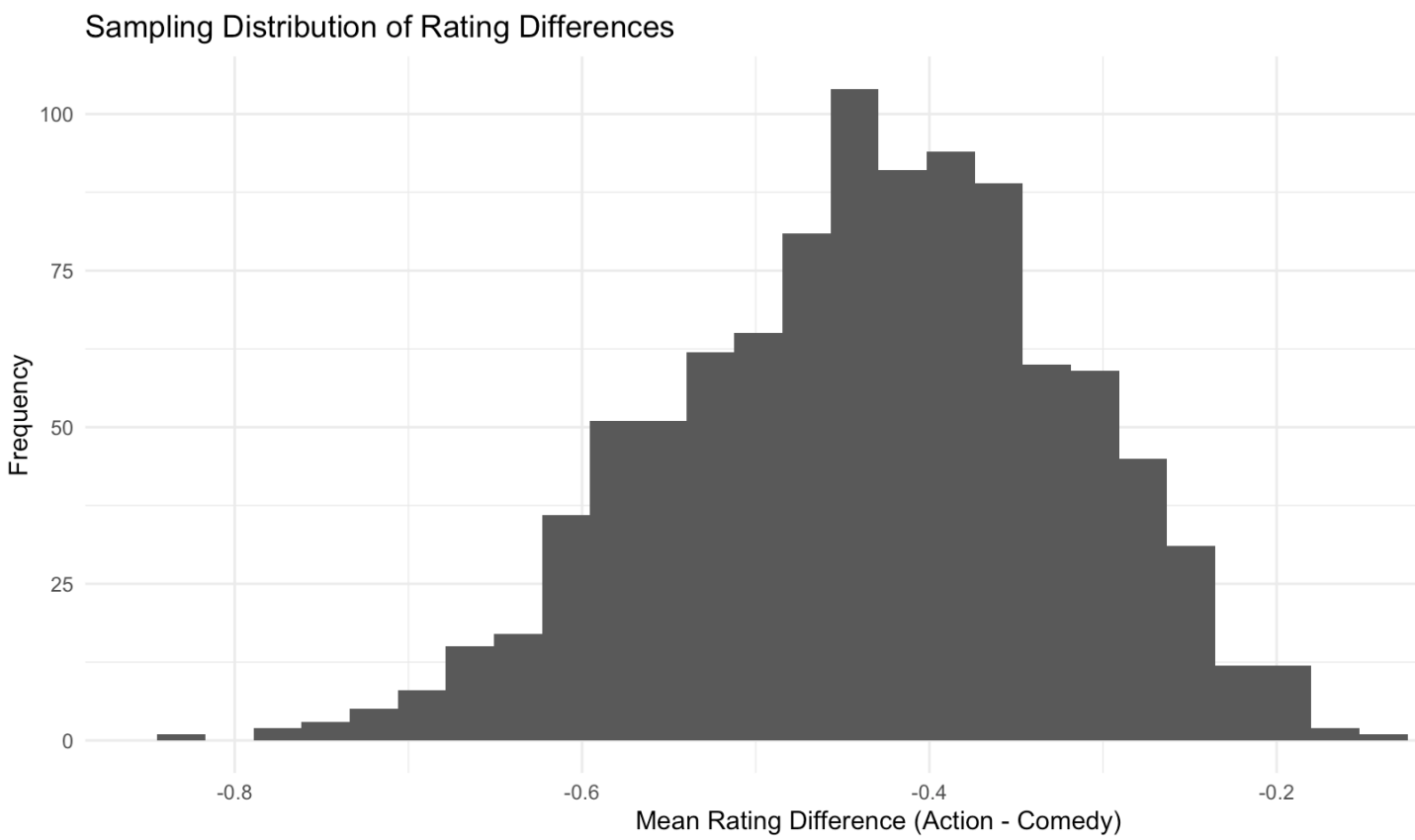
```r
# True effect world population
imaginary_movies_true <- tibble(
  movie_id = 1:1000000,
  genre = sample(c("Comedy", "Action"), size = 1000000, replace = TRUE),
  rating = ifelse(
    genre == "Comedy",
    rtruncnorm(1000000, a = 1, b = 10, mean = 6.0, sd = 2),
    rtruncnorm(1000000, a = 1, b = 10, mean = 5.5, sd = 2)
  )
)
```

2. Get your sampling distribution: Simulate 1000 random sa
store the results.

```
n_simulations <- 1000
differences <- c() # make an empty vector
sample_size <- 1000

data <- imaginary_movies_true # replace with imaginary_movies_null

for (i in 1:n_simulations) {
  # draw a sample of films
  imaginary_sample <- data |>
    sample_n(sample_size)
  # compute rating difference in the sample
  estimate <- imaginary_sample |>
    group_by(genre) |>
    summarize(avg_rating = mean(rating)) |>
    summarise(diff = avg_rating[genre == "Action"] - avg_rating[genre ==
    pull(diff)

  differences[i] <- estimate
}
```

3. Plot your Sampling distribution (use `data.frame()` to t
frame that can be read by `ggplot`)

```
ggplot(data.frame(differences), aes(x = differences)) +
  geom_histogram() +
  labs(title = "Sampling Distribution of Rating Differences",
       x = "Mean Rating Difference (Action - Comedy)",
       y = "Frequency") +
  theme_minimal()
```

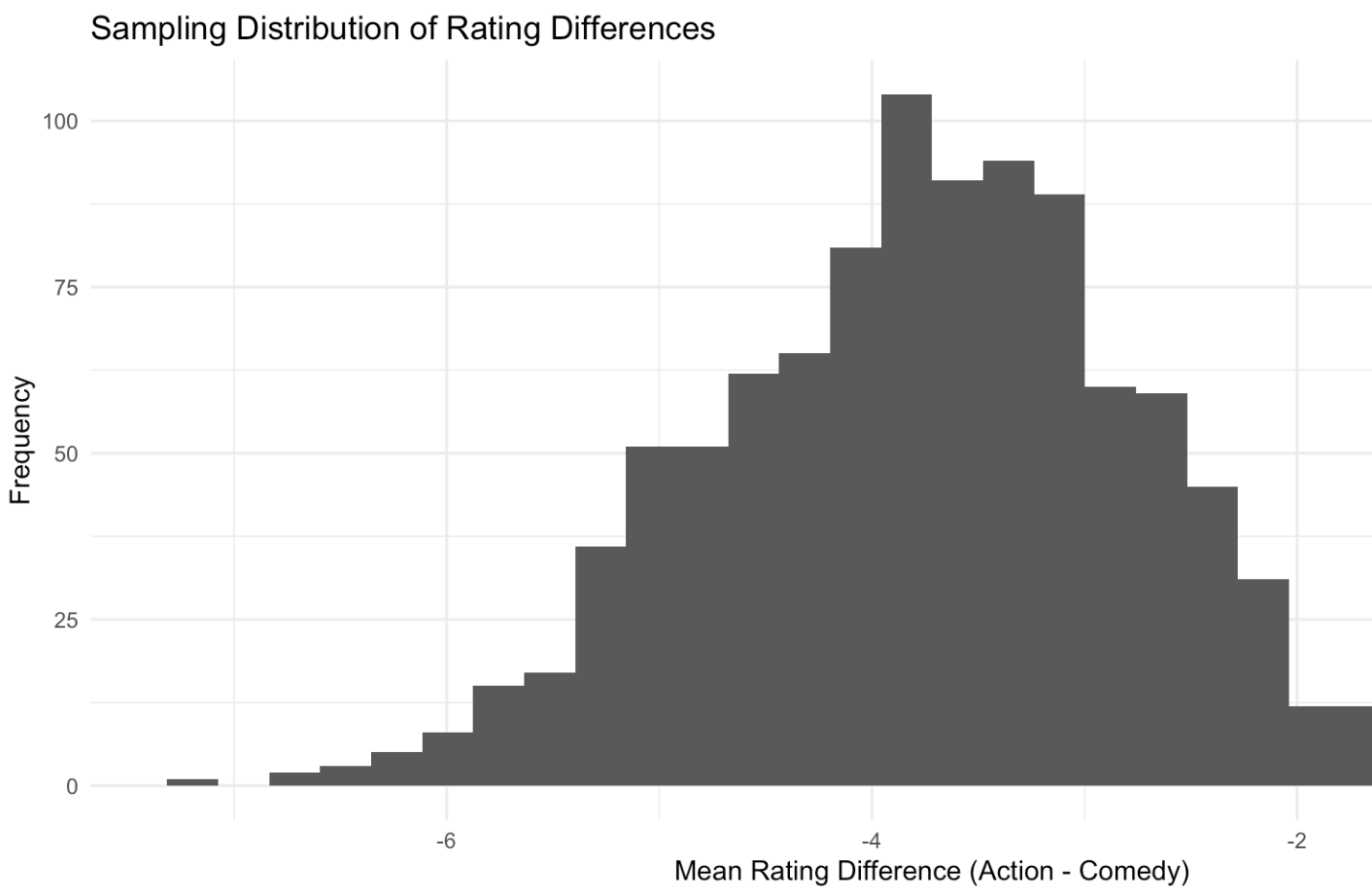


4. Prepare for hypothesis testing: bring the results on scale
distribution (divide by the standard deviation of your dist

```
differences_standardized <- differences/sd(differences)
```

5. Check the transformation: plot the new, standardized val

```
ggplot(data.frame(differences_standardized), aes(x = differences_standard
  geom_histogram() +
  labs(title = "Sampling Distribution of Rating Differences",
       x = "Mean Rating Difference (Action - Comedy)",
       y = "Frequency") +
  theme_minimal()
```



6. Calculate your power: Check how many (standardized) di
threshold, i.e. smaller than or equal to -1.96 (hint: use `mut`
with `ifelse` to create a new variable `significant` tha
or `FALSE`. Then use `summarize()` and `sum()` to calcula
above the 80% power threshold?

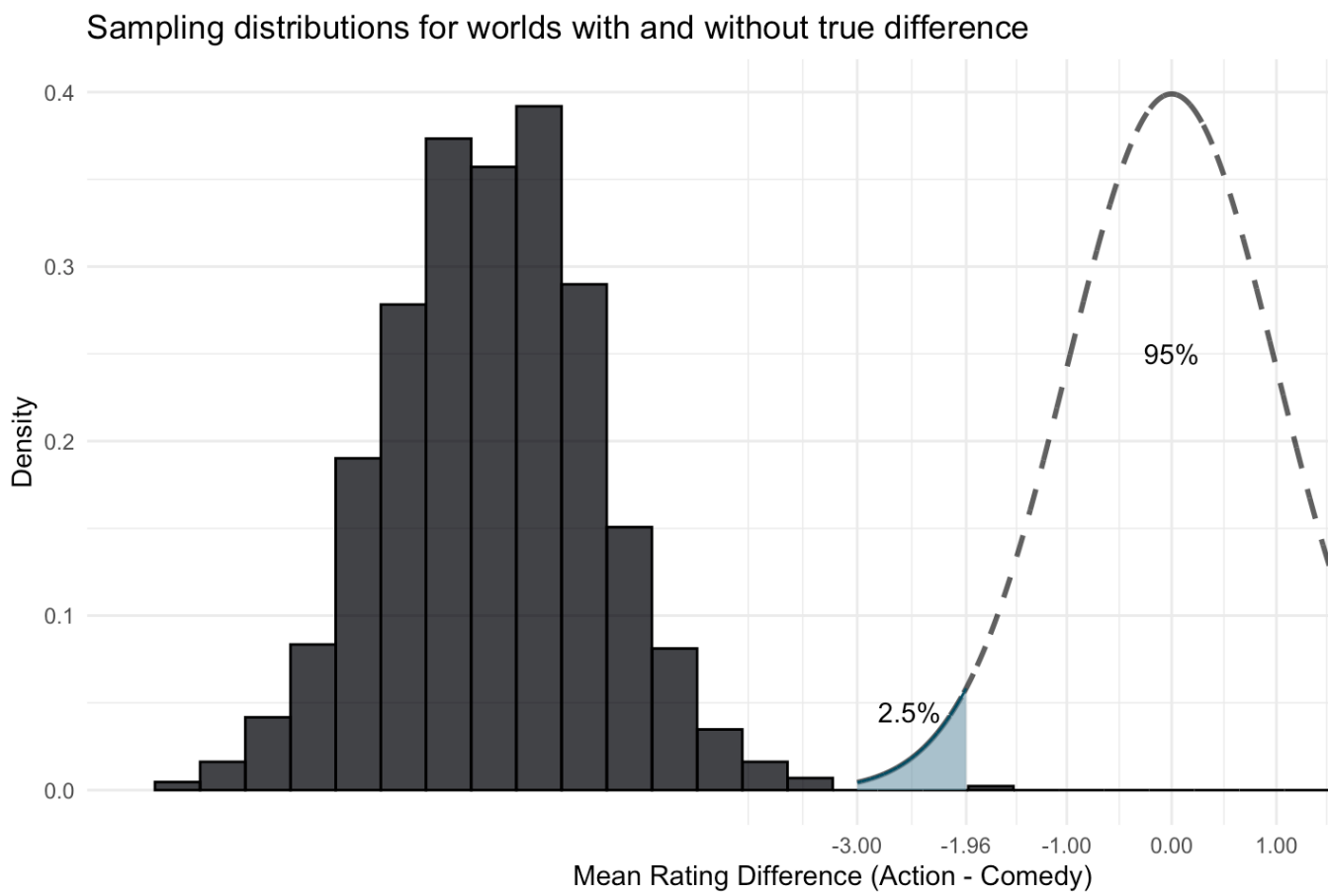
```
data.frame(differences_standardized) |>
  mutate(significant = ifelse(differences_standardized <= -1.96, TRUE, FA
  summarize(sum_significant = sum(significant),
            # you can also calculate the share directly
            share_significant = sum(significant) / n()
            )
```

```
  sum_significant share_significant
1             975             0.975
```

A power simulation

The central limit theorem (ag

The larger the sample size, the more statistical power



An example of a po
analysis

For an example, head over to the guide on power analysis on

That’s it for today :)